



Field Visit + Journey Tracking – Collaboration Document

◆ 1. Authentication

- **Method:** Token-based (**Bearer Token** with Laravel Sanctum / JWT).

Headers for all requests:

Authorization: Bearer <token>
Content-Type: application/json

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◆ 2. Flutter → Laravel API Parameters

A. User Login

- **Endpoint:** **POST** `/api/login`

Request:

```
{  
  "email": "user@example.com",  
  "password": "secret123"  
}
```

-

Response:

```
{  
  "token": "abcd1234xyz",  
}
```

```
"user": {  
  "id": 1,  
  "name": "John Doe",  
  "role": "employee"  
}
```

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B. Visit Check-In

- **Endpoint:** `POST /api/visits/checkin`

Request (Flutter must send):

```
{  
  "client_name": "ABC Retailer",  
  "latitude": 18.5204,  
  "longitude": 73.8567,  
  "notes": "Demo meeting scheduled"  
}
```

-

Response:

```
{  
  "success": true,  
  "visit": {  
    "id": 12,  
    "client_name": "ABC Retailer",  
    "check_in_time": "2025-09-26T09:15:00"  
  }  
}
```

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C. Visit Check-Out

- **Endpoint:** `POST /api/visits/{id}/checkout`

Request:

```
{  
  "notes": "Order confirmed, follow-up next week"  
}
```

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Response:

```
{  
  "success": true,  
  "visit": {  
    "id": 12,  
    "check_out_time": "2025-09-26T11:45:00"  
  }  
}
```

•

D. Location Logging (Background Journey Tracking)

- **Endpoint:** `POST /api/location/log`

Request (Flutter must send every 1–5 mins):

```
{  
  "latitude": 18.5211,  
  "longitude": 73.8595,  
  "accuracy": 15,  
  "battery": 82  
}
```

•

Response:

```
{  
  "success": true,
```

```
"log": {
  "id": 101,
  "latitude": 18.5211,
  "longitude": 73.8595,
  "logged_at": "2025-09-26T10:05:00"
}
}
```

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E. Journey Fetch (Per Day / Per Employee)

- **Endpoint:** `GET /api/location/journey?date=2025-09-26&user_id=3`

Response:

```
[
  { "latitude": 18.5204, "longitude": 73.8567, "logged_at": "2025-09-26T09:15:00" },
  { "latitude": 18.5301, "longitude": 73.8620, "logged_at": "2025-09-26T09:45:00" },
  { "latitude": 18.5405, "longitude": 73.8700, "logged_at": "2025-09-26T10:15:00" }
]
```

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F. Live Tracking (Latest Location of All Employees)

- **Endpoint:** `GET /api/location/live`

Response:

```
[
  {
    "user": { "id": 2, "name": "Rahul Singh" },
    "latitude": 18.5302,
    "longitude": 73.8600,
    "logged_at": "2025-09-26T10:20:00"
  },
  {
    "user": { "id": 3, "name": "Sneha Patel" },

```

```
"latitude": 18.5401,  
"longitude": 73.8650,  
"logged_at": "2025-09-26T10:22:00"  
}  
]
```

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◆ 3. Backend Handling (Laravel)

Middleware

- Use `auth:sanctum` or `jwt.auth` for all APIs.

Visit Handling

- On `checkin` → create visit with `check_in_time`.
- On `checkout` → update visit with `check_out_time`.
- Notes can be updated anytime.

Location Logs

- Store **lat, lng, accuracy, battery, timestamp**.
- Each entry is tied to `user_id`.
- Backend should **ignore duplicate points** (same lat/lng within 1 min).

Reports

- Daily journey → group by `user_id` + `date`.
- Live tracking → get latest log for each `user_id`.
- Distance traveled → Haversine formula between consecutive logs.

◆ 4. Flutter Developer Checklist

- Request **location permission** (foreground + background).
- Capture **GPS every 1–5 mins** (adjustable).
- Send **latitude, longitude, accuracy, battery** to API.
- Call **check-in / check-out API** manually via button.
- Store locally if offline → sync when online.
- Show journey history on map (Polyline).

◆ 5. Laravel Developer Checklist

- Create **migrations** for **visits** + **location_logs**.
- Secure APIs with **auth:sanctum**.
- Implement **checkin, checkout, log, journey, live** endpoints.
- Optimize journey queries (order by timestamp).
- Add API response standardization (**success, message, data**).
- Build manager dashboard (Leaflet / Google Maps).

👉 This document can act as the **API contract** between Flutter & Laravel teams.

Do you want me to **format this into a Google Docs / Notion template** so your team can directly collaborate (add comments, track changes)?